

Stat 6841 Homework 5

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Problem 1.

- (a) To simulate a standard Brownian motion using the method described, we use the code in the Python script `brownian.py`, reproduced in Listing 1. An example graph produced by the code is given in Figure 1. We observe that the paths with higher values of N have more fluctuations and smaller jumps.

Listing 1: Simulating a standard Brownian motion

```
1 import numpy as np
2
3 def simulate(n):
4     # Allocate empty array to store the path
5     b = np.empty(n+1)
6
7     # Generate evenly-spaced time points
8     t = np.linspace(0, 1, n+1)
9
10    # Initialize path to 0
11    b[0] = 0
12
13    # Generate path by adding N(0,1/sqrt(n)) random value
14    # to the previous value
15    for i in range(n):
16        b[i+1] = b[i] + np.random.normal()/n**.5
17
18    return t, b
19
20
21 if __name__ == '__main__':
22     import matplotlib.pyplot as plt
23
24     t50, b50 = simulate(50)
25     t100, b100 = simulate(100)
26     t1000, b1000 = simulate(1000)
27
28     plt.plot(t50, b50, label='N=50')
29     plt.plot(t100, b100, label='N=100')
30     plt.plot(t1000, b1000, label='N=1000')
31     plt.legend()
32     plt.show()
```

- (b) Let $B(t)$ be a standard Brownian motion. Then $D(t) = \mu t + \sigma B(t)$ is a Brownian motion with drift μ and standard deviation σ . Thus, given a simulated path $\{b(0), b(t_1), \dots, b(t_N)\}$ of $\{B(t)\}$, the sequence

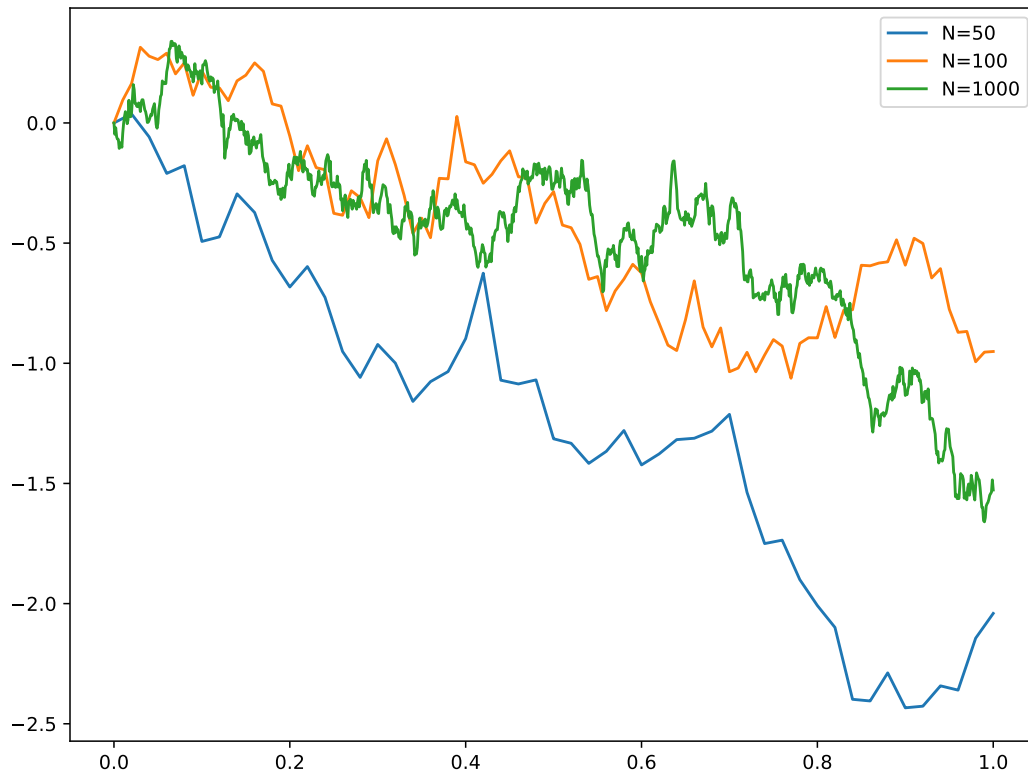


Figure 1: Three simulated paths of a standard Brownian motion.

$\{\mu \cdot 0 + \sigma b(0), \mu t_1 + \sigma b(1), \dots, \mu t_N + \sigma b(t_N)\}$ is a simulated path of $\{D(t)\}$. This is implemented in `drift.py`, which is reproduced in Listing 2. Three sample simulated paths are shown in Figure 2.

Listing 2: Simulating a Brownian motion with drift and standard deviation

```

1 import brownian
2
3 def simulate(n, mu, sigma):
4     # Generate standard motion
5     t, b = brownian.simulate(n)
6
7     # Transform to motion with drift + standard deviation
8     return t, mu * t + sigma * b
9
10
11 if __name__ == '__main__':
12     import matplotlib.pyplot as plt
13 
```

```

14     t50, b50 = simulate(50, mu=2, sigma=0.05)
15     t100, b100 = simulate(100, mu=2, sigma=0.05)
16     t1000, b1000 = simulate(1000, mu=2, sigma=0.05)
17
18     plt.plot(t50, b50, label='N=50 ')
19     plt.plot(t100, b100, label='N=100 ')
20     plt.plot(t1000, b1000, label='N=1000 ')
21     plt.legend()
22     plt.show()

```

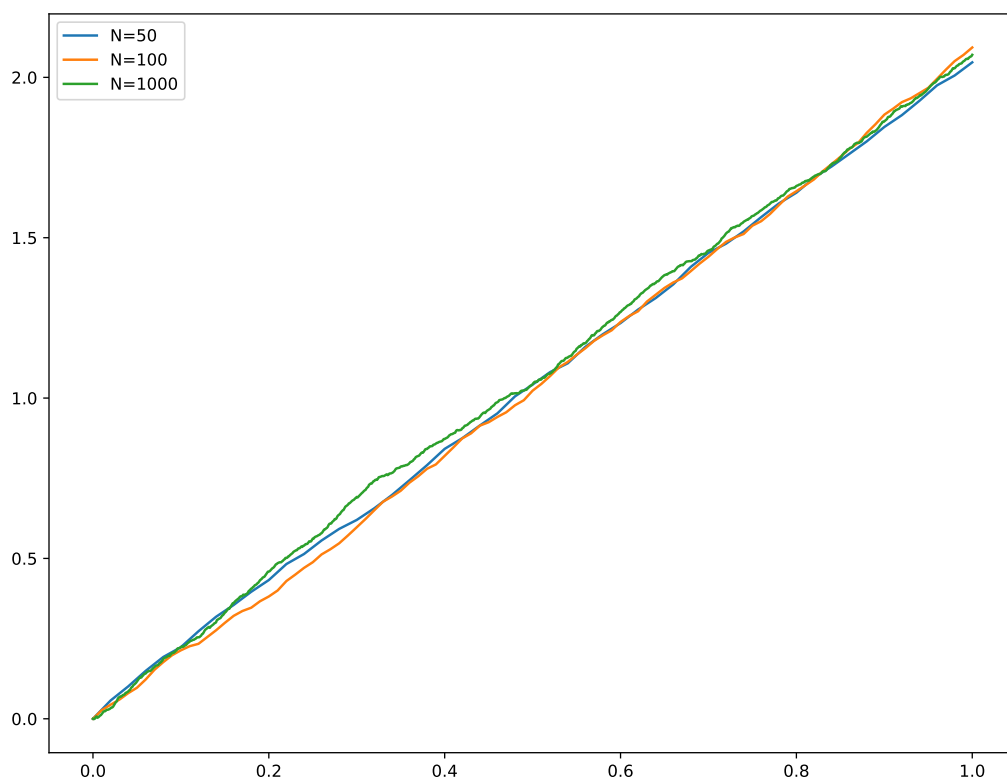


Figure 2: Three simulated paths of a Brownian motion with drift $\mu = 2$ and standard deviation $\sigma = 0.05$.

(c) Let $B(t)$ be a standard Brownian motion. We simulate the Ornstein-Uhlenbeck process

$$X(t) = e^{-t} B(e^{2t}).$$

We would like to simulate a sample path $\{x(0), x(t_1), \dots, x(t_N)\}$ for this process. Since we can write

$$B(e^{2t_n}) = (B(e^{2t_n}) - B(e^{2t_{n-1}})) + B(e^{2t_{n-1}})$$

as a sum of the independent random variables $(B(e^{2t_n}) - B(e^{2t_{n-1}}))$ and $B(e^{2t_{n-1}})$, we can simulate $\{B(e^0), B(e^{2t_1}), \dots, B(e^{2t_N})\}$ by simulating $B(e^{2t_{n-1}})$, then simulating the difference $B(e^{2t_n}) - B(e^{2t_{n-1}}) \sim N(0, e^{2t_n} - e^{2t_{n-1}})$ and adding it to the simulated value of $B(e^{2t_{n-1}})$. This is implemented in Listing 3. Three sample simulated paths are shown in Figure 3.

Listing 3: Simulating an Ornstein-Uhlenbeck process

```

1 import numpy as np
2
3
4 def simulate(n):
5     # Get t values
6     t = np.linspace(0, 1, n+1)
7
8     # Initialize process x and B(e^{2*0}) = B(1) ~ N(0,1)
9     x = np.empty(n+1)
10    b = np.random.normal()
11    x[0] = b
12
13    for i in range(n):
14        # Compute B(e^{2t_{i+1}})
15        b += np.random.normal() * (np.exp(2*t[i+1]) - np.exp(2*t[i]))
16        x[i+1] = b * np.exp(-t[i+1])
17
18    return t, x
19
20
21 if __name__ == '__main__':
22     import matplotlib.pyplot as plt
23
24     t50, x50 = simulate(50)
25     t100, x100 = simulate(100)
26     t1000, x1000 = simulate(1000)
27
28     plt.plot(t50, x50, label='N=50')
29     plt.plot(t100, x100, label='N=100')
30     plt.plot(t1000, x1000, label='N=1000')
31     plt.legend()
32     plt.show()

```

Problem 2.

Let Φ be the CDF of a standard normal random variable.

- (a) Since $B(2) \sim N(0, 2)$, it follows that $Z = \frac{1}{\sqrt{2}}B(2)$ follows the distribution $N(0, 1)$. Then

$$P(B(2) \leq 1) = P\left(\sqrt{2}Z \leq 1\right) = P\left(Z \leq \frac{1}{\sqrt{2}}\right) = \Phi\left(\frac{1}{\sqrt{2}}\right) \approx 0.760.$$

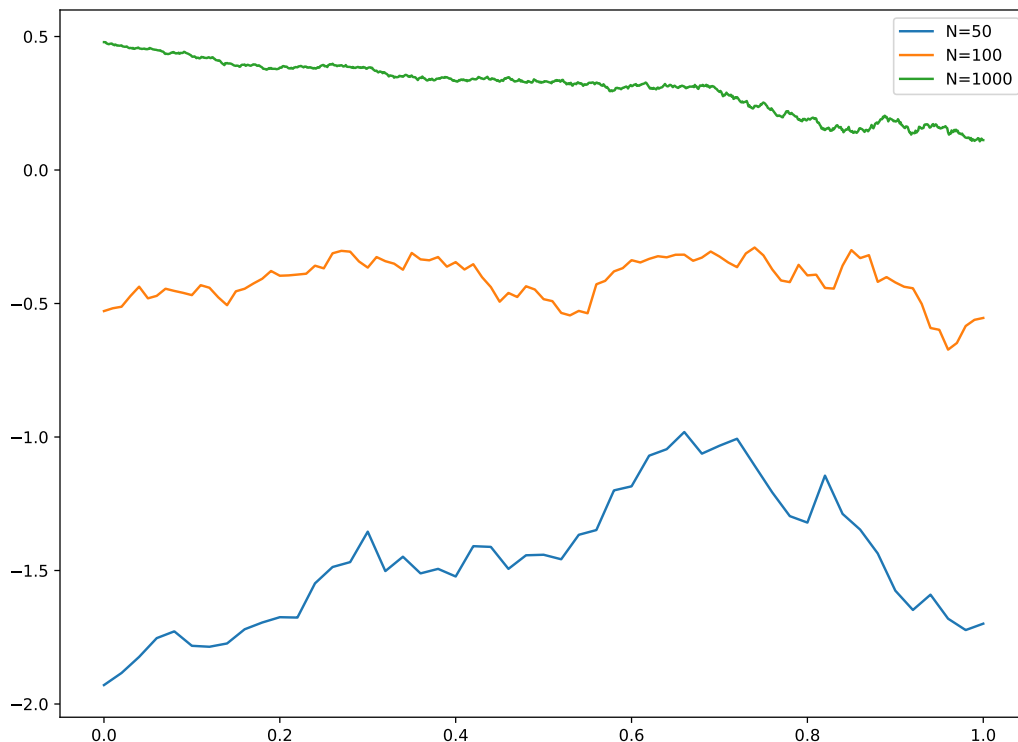


Figure 3: Three simulated paths of a standard Ornstein-Uhlenbeck process.

(b) By the independent and identical increments property and the fact that $B(3) \sim N(0, 3)$, we have

$$\begin{aligned}
 E[B(4)|B(1) = x] &= E[B(4) - B(1)|B(1) = x] + E[B(1)|B(1) = x] \\
 &= E[B(4) - B(1)|B(1) - B(0) = x] + x \\
 &= E[B(4) - B(1)] + x \\
 &= E[B(3)] + x \\
 &= x.
 \end{aligned}$$

(c) By the independent and identical increments property and the fact that $B(t) \sim N(0, t)$ for all t , we have

$$\begin{aligned}
 \text{Var}[B(4)|B(1)] &= \text{Var}[B(4) - B(1)|B(1) - B(0)] + \text{Var}[B(1)|B(1)] \\
 &= \text{Var}[B(4) - B(1)] + E[(B(1) - E[B(1)|B(1)])^2|B(1)] \\
 &= \text{Var}[B(3)] \\
 &= 3.
 \end{aligned}$$

(d) By the independent and identical increments property, we have

$$P(B(3) \leq 5|B(1) = 2) = P(B(3) - B(1) \leq 3|B(1) - B(0) = 2) = P(B(3) - B(1) \leq 3) = P(B(2) \leq 3).$$

Since $B(2) \sim N(0, 2)$, it follows that $Z = \frac{1}{\sqrt{2}}B(2)$ is a standard normal random variable. Then

$$P(B(3) \leq 5 | B(1) = 2) = P(B(2) \leq 3) = P(\sqrt{2}Z \leq 3) = P\left(Z \leq \frac{3}{\sqrt{2}}\right) = \Phi\left(\frac{3}{\sqrt{2}}\right) \approx 0.983.$$

Problem 3.

1. We have

$$X(0) = (1+0)U\left(\frac{0}{1+0}\right) = U(0) = B(0) - 0 \cdot B(1) = 0.$$

2. Let $0 \leq r < s < t$. Then

$$\begin{aligned} X(t) - X(s) &= (1+t)U\left(\frac{t}{1+t}\right) - (1+s)U\left(\frac{s}{1+s}\right) \\ &= (1+t)B\left(\frac{t}{1+t}\right) - tB(1) - (1+s)B\left(\frac{s}{1+s}\right) - sB(1) \end{aligned}$$

Problem 4.

(a) We know that $(B(s), B(t), B(u))$ is multivariate normal with mean zero and covariance matrix

$$\Sigma = \begin{bmatrix} s & s & s \\ s & t & t \\ s & t & u \end{bmatrix} = \begin{bmatrix} A & C \\ C^T & u \end{bmatrix}, \quad A = \begin{bmatrix} s & s \\ s & t \end{bmatrix}, \quad C = \begin{bmatrix} s \\ t \end{bmatrix}.$$

Then $X = (B(s), B(t)) | B(u)$ is also multivariate normal with

$$E[X] = Cu^{-1}B(u), \quad \text{Cov}(X) = \Sigma/u = A - Cu^{-1}C^T,$$

where Σ/u is the Schur complement. Then

$$E[X] = \begin{bmatrix} \frac{s}{u} \\ \frac{t}{u} \end{bmatrix} B(u), \quad \text{Cov}(X) = \begin{bmatrix} s & s \\ s & t \end{bmatrix} - \frac{1}{u} \begin{bmatrix} s^2 & st \\ st & t^2 \end{bmatrix}.$$

Thus,

$$\begin{aligned}
E[B(s)B(t)|B(u)] &= \text{Cov}(B(s), B(t)|B(u)) + E[B(s)|B(u)]E[B(t)|B(u)] \\
&= [\text{Cov}(X)]_{12} + [E[X]]_1[E[X]]_2 \\
&= s - \frac{st}{u} + \frac{s}{u} \cdot \frac{t}{u} \cdot (B(u))^2 \\
&= s - \frac{st}{u} + \frac{s}{t}[E[X]]_1^2 \\
&= s - \frac{st}{u} + \frac{s}{t}E[B(t)|B(u)]^2 \\
&= s - \frac{st}{u} + \frac{s}{t}(E[(B(t))^2|B(u)] - \text{Var}[B(t)|B(u)]) \\
&= s - \frac{st}{u} + \frac{s}{t}(E[(B(t))^2|B(u)] - [\text{Cov}(X)]_{22}) \\
&= s - \frac{st}{u} + \frac{s}{t}\left(E[(B(t))^2|B(u)] - t + \frac{t^2}{u}\right) \\
&= \frac{s}{t}E[(B(t))^2|B(u)].
\end{aligned}$$

(b) Using the third line and the last line in the final chain of equations in part (a), we have

$$\frac{s}{t}E[(B(t))^2|B(u)] = s - \frac{st}{u} + \frac{st}{u^2}(B(u))^2,$$

which implies that

$$E[(B(t))^2|B(u)] = t - \frac{t^2}{u} + \frac{t^2}{u^2}(B(u))^2.$$

Problem 5.

We know that $(B(5), B(9), B(12), B(10))$ is multivariate normal with mean zero and covariance matrix

$$\Sigma = \begin{bmatrix} 5 & 5 & 5 & 5 \\ 5 & 9 & 9 & 9 \\ 5 & 9 & 12 & 10 \\ 5 & 9 & 10 & 10 \end{bmatrix} = \begin{bmatrix} A & C \\ C^T & 10 \end{bmatrix}, \quad A = \begin{bmatrix} 5 & 5 & 5 \\ 5 & 9 & 9 \\ 5 & 9 & 12 \end{bmatrix}, \quad C = \begin{bmatrix} 5 \\ 9 \\ 10 \end{bmatrix}.$$

Then $X = (B(5), B(9), B(12))|B(10)$ is also multivariate normal with

$$E[X] = C10^{-1}B(10), \quad \text{Cov}(X) = \Sigma/10 = A - C10^{-1}C^T,$$

where $\Sigma/10$ is the Schur complement. Thus, given $B(10) = 3$, we have

$$E[X] = \frac{3}{10} \begin{bmatrix} 5 \\ 9 \\ 10 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 2.7 \\ 3 \end{bmatrix}, \quad \text{Cov}(X) = A - \frac{1}{10} \begin{bmatrix} 25 & 45 & 50 \\ 45 & 81 & 90 \\ 50 & 90 & 100 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 25 & 5 & 0 \\ 5 & 9 & 0 \\ 0 & 0 & 20 \end{bmatrix} = \begin{bmatrix} 2.5 & 0.5 & 0 \\ 0.5 & 0.9 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Problem 6.

(a) For $a > 0$, let $T_a = \inf\{t \mid B(t) \geq a\}$. Then

$$P(B(t) \geq a) = P(B(t) \geq a | T_a \leq t)P(T_a \leq t) + P(B(t) \geq a | T_a > t)P(T_a > t).$$

Since $B(t) \geq a$ implies that $B(s) = a$ for some $s \leq t$ by the continuity of $B(t)$, it follows that $T_a > t$ and $B(t) \geq a$ are disjoint. Then

$$P(B(t) \geq a) = P(B(t) \geq a | T_a \leq t)P(T_a \leq t).$$

If $T_a \leq t$, then $B(t) - B(T_a) = B(t) - a$ should be equally likely to be positive or negative, which implies that $P(B(t) \geq a | T_a \leq t) = \frac{1}{2}$. Then

$$P(T_a \leq t) = 2P(B(t) \geq a).$$

We now observe that $X(t) \leq a$ if and only if $T_a \geq t$, for $a > 0$. Furthermore, $P(X(t) \leq 0) = 0$. Thus, the CDF of $X(t)$ is given by

$$F(x) = P(X(t) \leq x) = 1 - 2P(B(t) \geq x) = 2P(B(t) \leq x) - 1, \quad x \geq 0.$$

Since $B(t) \sim N(0, t)$, it follows that

$$F(x) = 2\Phi\left(\frac{x}{\sqrt{t}}\right) - 1, \quad x \geq 0,$$

where Φ is the CDF of the standard normal distribution. Then the PDF of $X(t)$ is given by

$$f(x) = F'(x) = 2\phi\left(\frac{x}{\sqrt{t}}\right) \frac{1}{\sqrt{t}} = \frac{2}{\sqrt{t}} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \frac{x^2}{t}}, \quad x \geq 0,$$

where ϕ is the PDF of the standard normal distribution.

(b) We have

$$\begin{aligned} E[X(t)] &= \frac{2}{\sqrt{t}} \frac{1}{\sqrt{2\pi}} \int_0^\infty x e^{-\frac{1}{2} \frac{x^2}{t}} dx \\ &= \frac{2}{\sqrt{t}} \frac{1}{\sqrt{2\pi}} t \int_0^\infty e^{-u} du \\ &= \sqrt{2t\pi^{-1}}. \end{aligned}$$

Furthermore,

$$\begin{aligned} E[(X(t))^2] &= \frac{2}{\sqrt{t}} \frac{1}{\sqrt{2\pi}} \int_0^\infty x^2 e^{-\frac{1}{2} \frac{x^2}{t}} dx \\ &= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^\infty x^2 e^{-\frac{1}{2} \frac{x^2}{t}} dx \\ &= E[Y^2] \end{aligned}$$

where $Y \sim N(0, t)$. Since $E[Y^2] = \text{Var}[Y] + (E[Y])^2 = \text{Var}[Y] = t$, we get $E[(X(t))^2] = t$. Then

$$\text{Var}[X(t)] = E[(X(t))^2] - (E[X(t)])^2 = t - 2t\pi^{-1} = t(1 - 2\pi^{-1}).$$

7. Textbook: 3, p. 686

By the independent and identical increments property, we have

$$\begin{aligned} E[B(t_1)B(t_2)B(t_3)] &= E[B(t_1)(B(t_2) - B(t_1))(B(t_3) - B(t_2)) + (B(t_1))^2(B(t_3) - B(t_2)) + B(t_1)(B(t_2))^2] \\ &= E[B(t_1)(B(t_2) - B(t_1))^2 + 2(B(t_1))^2B(t_2) - (B(t_1))^3] \\ &= E[2(B(t_1))^2(B(t_2) - B(t_1)) + (B(t_1))^3] \\ &= E[(B(t_1))^3] = 0 \end{aligned}$$

because the third moment of a random variable with distribution $N(\mu, \sigma)$ is $\mu^3 + 3\mu\sigma^2$, and $B(t_1) \sim N(0, t_1)$.

8. Textbook: 4, p. 686

We know from problem 6 (a) that the CDF of T_a is given by

$$F(t) = P(T_a \leq t) = 2P(B(t) \geq a) = 2(1 - P(B(t) \leq a)) = 2\left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right), \quad t > 0.$$

where Φ is the CDF of the standard normal distribution, as $B(t) \sim N(0, t)$. Then the PDF is given by

$$f(t) = \phi\left(\frac{a}{\sqrt{t}}\right) t^{-\frac{3}{2}}, \quad t > 0.$$

Then

$$P(T_a < \infty) = \int_0^\infty f(t) dt = \lim_{b \rightarrow \infty} \lim_{a \rightarrow 0^+} \int_a^b f(t) dt = \lim_{b \rightarrow \infty} \lim_{a \rightarrow 0^+} (F(b) - F(a)) = 1.$$

On the other hand,

$$E[T_a] = \int_0^\infty t f(t) dt = \int_0^\infty t^{-\frac{1}{2}} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \frac{a^2}{t}} dt.$$

Using the substituting $u = t^{-1}$, we have $-u^{-2} du = dt$, and

$$E[T_a] = \frac{1}{\sqrt{2\pi}} \int_0^\infty u^{-\frac{3}{2}} e^{-\frac{1}{2} a^2 u} du \geq \frac{e^{-\frac{1}{2} a^2}}{\sqrt{2\pi}} \int_0^1 u^{-\frac{3}{2}} du = \frac{e^{-\frac{1}{2} a^2}}{\sqrt{2\pi}} \lim_{t \rightarrow 0^+} \left[-2u^{-\frac{1}{2}}\right]_t^1 = \infty$$

because

$$\lim_{t \rightarrow 0^+} \left[-2u^{-\frac{1}{2}}\right]_t^1 = \lim_{t \rightarrow 0^+} \left(\frac{2}{\sqrt{t}} - 2\right) = \infty.$$

9. Textbook: 7, p. 686

Let $C(s) = B(s + t_1) - B(t_1)$ for $s > 0$. Then $\{C(s)\}$ is a standard Brownian motion because it satisfies the four properties.

1. We have $C(0) = B(t_1) - B(t_1) = 0$.
2. Given $s < t < u$, we have $C(u) - C(t) = B(u + t_1) - B(t + t_1)$, and $C(t) - C(s) = B(t + t_1) - B(s + t_1)$, which are independent because $\{B(t)\}$ is a Brownian motion. The stationarity of increments follows by similar reasoning.
3. By the stationarity of the increments of $\{B(t)\}$, it follows that $C(s) = B(s + t_1) - B(t_1) \sim N(0, s + t_1 - t_1) = N(0, s)$.
4. Since the paths of $\{B(t)\}$ are continuous, the paths of $\{C(s)\}$ are also continuous, being only offset from paths of $\{B(t)\}$ by a random constant $B(t_1)$.

We also note that $C(s)$ is independent of $B(t_1)$ for all $s > 0$. Hence, by the continuity of the paths of $\{B(s)\}$,

$$X = \max_{0 \leq s \leq t_2 - t_1} C(s) = \sup_{0 < s \leq t_2 - t_1} C(s)$$

is independent of $B(t_1)$. Then the PDF of

$$\max_{t_1 \leq s \leq t_2} B(s) = \max_{t_1 \leq s \leq t_2} (C(s - t_1) + B(t_1)) = B(t_1) + \max_{0 \leq s \leq t_2 - t_1} C(s) = B(t_1) + X$$

is given by the convolution

$$f = f_1 * f_2,$$

where f_1 is the PDF of $B(t_1)$, which is known because $B(t_1) \sim N(0, t_1)$, and f_2 is the PDF of X , which we derived in **6 (a)**. That is,

$$\begin{aligned} f_1(y) &= \frac{1}{\sqrt{2\pi t_1}} e^{-\frac{1}{2} \frac{y^2}{t_1}} \\ f_2(y) &= \frac{2}{\sqrt{t_2 - t_1}} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \frac{y^2}{t_2 - t_1}}, \quad y \geq 0. \end{aligned}$$

Then

$$\begin{aligned} f(y) &= (f_1 * f_2)(y) = \int_{-\infty}^{\infty} f_1(y - z) f_2(z) \, dz \\ &= \frac{1}{\pi \sqrt{t_1(t_2 - t_1)}} \int_0^{\infty} e^{-\frac{1}{2} \frac{(y-z)^2}{t_1}} e^{-\frac{1}{2} \frac{z^2}{t_2 - t_1}} \, dz \end{aligned}$$

because $f_2(z) = 0$ for $z < 0$. Finally, we can compute

$$\begin{aligned} P\left(\max_{t_1 \leq s \leq t_2} B(s) > x\right) &= P(B(t_1) + X > x) = \int_x^{\infty} f(y) \, dy \\ &= \frac{1}{\pi \sqrt{t_1(t_2 - t_1)}} \int_x^{\infty} \int_0^{\infty} e^{-\frac{1}{2} \frac{(y-z)^2}{t_1}} e^{-\frac{1}{2} \frac{z^2}{t_2 - t_1}} \, dz \, dy. \end{aligned}$$

10. Textbook: 10, p. 687

We note that $(X(s), X(t))$ is a bivariate normal distribution because $\{X(t)\}$ is a Gaussian process. The parameters of $(X(s), X(t))$ are

$$\boldsymbol{\mu} = \begin{bmatrix} E[X(s)] \\ E[X(t)] \end{bmatrix} = \mu \begin{bmatrix} s \\ t \end{bmatrix},$$

and

$$\Sigma = \begin{bmatrix} \text{Var}[X(s)] & \text{Cov}(X(s), X(t)) \\ \text{Cov}(X(s), X(t)) & \text{Var}[X(t)] \end{bmatrix} = \begin{bmatrix} \sigma^2 s & \text{Cov}(X(s), X(t)) \\ \text{Cov}(X(s), X(t)) & \sigma^2 t \end{bmatrix}.$$

Let $s < t$. Then

$$\text{Cov}(X(s), X(t)) = \text{Cov}(X(s), X(t) - X(s)) + \text{Cov}(X(s), X(s)) = \text{Var}[X(s)] = \sigma^2 s$$

because $X(t) - X(s)$ is independent of $X(s)$. Then, in general, $\text{Cov}(X(s), X(t)) = \sigma^2(s \wedge t)$. Thus,

$$\Sigma = \sigma^2 \begin{bmatrix} s & s \wedge t \\ s \wedge t & t \end{bmatrix}.$$

Finally, we use the fact that the conditional distribution $X(s)|(X(t) = c)$ is normal with

$$\begin{aligned} E[X(s)|X(t) = c] &= E[X(s)] + (\sigma^2(s \wedge t))(\sigma^2 t)^{-1}(c - E[X(t)]) = \mu s + \frac{s \wedge t}{t}(c - \mu t), \\ \text{Var}[X(s)|X(t) = c] &= \Sigma/t = \sigma^2 \left(s - (s \wedge t)t^{-1}(s \wedge t)\right) = \sigma^2 \left(s - \frac{(s \wedge t)^2}{t}\right), \end{aligned}$$

where Σ/t is the Schur complement.

(a) For $s < t$, we have

$$X(s)|(X(t) = c) \sim N\left(\frac{s}{t}c, \sigma^2 s \left(1 - \frac{s}{t}\right)\right).$$

(b) For $t < s$, we have

$$X(s)|(X(t) = c) \sim N(\mu(s - t) + c, \sigma^2(s - t)).$$

11. Textbook: 17, p. 687

Let $\mathcal{F}_t = \sigma(\{B(s) : s \leq t\})$. Then $\{B(t)\}$ is a martingale with respect to $\{\mathcal{F}_t\}$ because it satisfies all three requirements.

1. Certainly $\{B(t)\}$ is adapted to $\{\mathcal{F}_t\}$ by construction.

2. For any $t > 0$, we have

$$E[|B(t)|] = \int_{-\infty}^{\infty} |x| \frac{1}{\sqrt{2\pi t}} e^{-\frac{1}{2} \frac{x^2}{t}} dx = \frac{2}{\sqrt{2\pi t}} \int_0^{\infty} x e^{-\frac{1}{2} \frac{x^2}{t}} dx < \infty,$$

which can be seen easily by making the substitution $u = x^2$.

3. Let $s < t$. Then by the independence of increments, $B(t) - B(s)$ is independent of $\mathcal{F}_s$, and

$$E[B(t)|\mathcal{F}_s] = E[B(t) - B(s)|\mathcal{F}_s] + E[B(s)|\mathcal{F}_s] = E[B(t)] - E[B(s)] + B(s) = B(s).$$

12. Textbook: 18, p. 687

Let $\mathcal{F}_t = \sigma(\{B(s) : s \leq t\})$. Then $\{Y(t)\}$ is a martingale with respect to $\{\mathcal{F}_t\}$ because it satisfies all three requirements.

1. Since $Y(t) = B^2(t) - t$ is a continuous function of $B(t)$, which is adapted to $\mathcal{F}_t$ for all t , it follows that $\{Y(t)\}$ is adapted to $\{\mathcal{F}_t\}$.

2. Let $t > 0$. Then

$$E[Y(t)] = E[B^2(t)] - t = 0 < \infty$$

because $B(t) \sim N(0, t)$.

3. Let $0 < s < t$. Using the independent and identical increments property, we have

$$\begin{aligned} E[Y(t)|\mathcal{F}_s] &= E[B^2(t) - t|\mathcal{F}_s] \\ &= E[(B(t) - B(s))^2 + 2(B(t) - B(s))B(s) + B^2(s)|\mathcal{F}_s] - t \\ &= E[(B(t) - B(s))^2] + 2B(s)E[B(t) - B(s)] + B^2(s) - t \\ &= t - s + B^2(s) - t \\ &= Y(s). \end{aligned}$$